CSE4104 | PROJECT PROPOSAL

EduNexa AI

Your Intelligent Learning & Project Workspace

Department of CSE, Northern University of Business and Technology, Khulna

Field	Details
Official Team Name	CSE4104 -7E -T05
Section	7E
Project Title	EduNexa AI: Your Intelligent Learning & Project Workspace
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1. Project Title

EduNexa AI: Your Intelligent Learning & Project Workspace

EduNexa AI is an AI-based web platform where university students turn their own study material into flashcards, notes, quizzes and cited answers, and get structured help with their course projects.

2. Problem Statement

2.1 What problem currently exists

Every semester, university students collect a large amount of study material: lecture slides, PDF books, class notes and video lectures. Turning this material into revision notes, flashcards and practice questions takes many hours, especially before exams. Course projects add more pressure, because students also have to choose a topic, plan the work and write a report, often with very little guidance.

2.2 Why the problem is important

- Time spent copying and rewriting material is time not spent understanding it, so exam preparation becomes stressful and shallow.
- Students cannot easily see how much they have studied or how regular their study habit is.

- A weak start in a course project (a poor topic or an unclear plan) affects the quality of the whole semester's work.
- Good study support is not equally available to all students, especially where tools must be paid for in foreign currency, as in Bangladesh.

2.3 Why existing solutions are insufficient

- General chatbots do not know a student's own course material, so their answers are often unrelated to the syllabus and may contain made-up facts.
- Flashcards, notes, quizzes and video summaries are spread over separate apps, so students waste time switching between them.
- Most good AI study tools are paid and priced in dollars, which many students cannot afford.
- Few tools help with the early stages of a course project, such as idea selection, outlining and report planning.
- Many study apps have dense, dark and crowded interfaces that feel tiring and intimidating for long use.

2.4 Why AI is relevant

The task is to read long, unstructured text and produce new, useful text from it. Fixed rules cannot do this, but language models can summarise content, write questions and explain concepts step by step. With Retrieval-Augmented Generation (RAG), the answers are based on the student's own documents instead of the model's general memory, and each answer can show its source. Embeddings allow search by meaning, not only by keyword, and vision models can read scanned pages and images. AI is therefore the core of the solution, not an add-on.

3. Objectives

- Generate flashcards and quizzes automatically from text, a PDF or a topic name.
- Let students chat with a PDF or a YouTube video and receive answers based on the actual content, with source references.
- Create short, structured notes and step-by-step answers from text or images.
- Provide an AI Tutor for concept explanations and a Project Assistant for project ideas, outlines and report drafting help.
- Show learning progress, study streak and recent activity on a simple dashboard to improve

student productivity.

- Offer a calm, easy-to-use interface with a soft light theme, an optional dark mode and a mobile-friendly layout.
- Provide Free and Pro plans with usage limits, so the service stays affordable and sustainable.

4. Proposed Features :

Feature	What it does	Role of AI
Authentication	Sign up and login with email or Google; private data for each user.	None
Dashboard	Greeting, quick actions, progress ring, study streak, recent activity and popular resources.	None
Flashcard Generator	Creates question-and-answer cards from notes, text or a topic. Cards flip and show progress.	Generates the cards
Chat with PDF	The student uploads a PDF and asks questions. Answers come from the document, with page references.	RAG, embeddings, answer generation
Chat with YouTube	The video transcript is read and turned into a summary, key points and answers.	RAG, summarisation
Magic Notes	Makes short, structured notes from a PDF, a web page or an image.	Summarisation, vision model
Quiz Generator	Makes multiple-choice questions with scoring so students can test themselves.	Generates questions and options
AI Tutor	Explains difficult concepts in simple steps and helps with problem solving.	Conversational explanation
Instant Answers	Gives step-by-step answers to any question, shown as they are written.	Streaming answer generation

Project Assistant	Suggests project ideas, builds an outline and helps with report writing.	Idea and outline generation
Progress Tracking	Records activity to show study streak and recent work.	None
Plans and Usage Limits	Free and Pro plans, daily limits and a sandbox payment flow for upgrade.	None

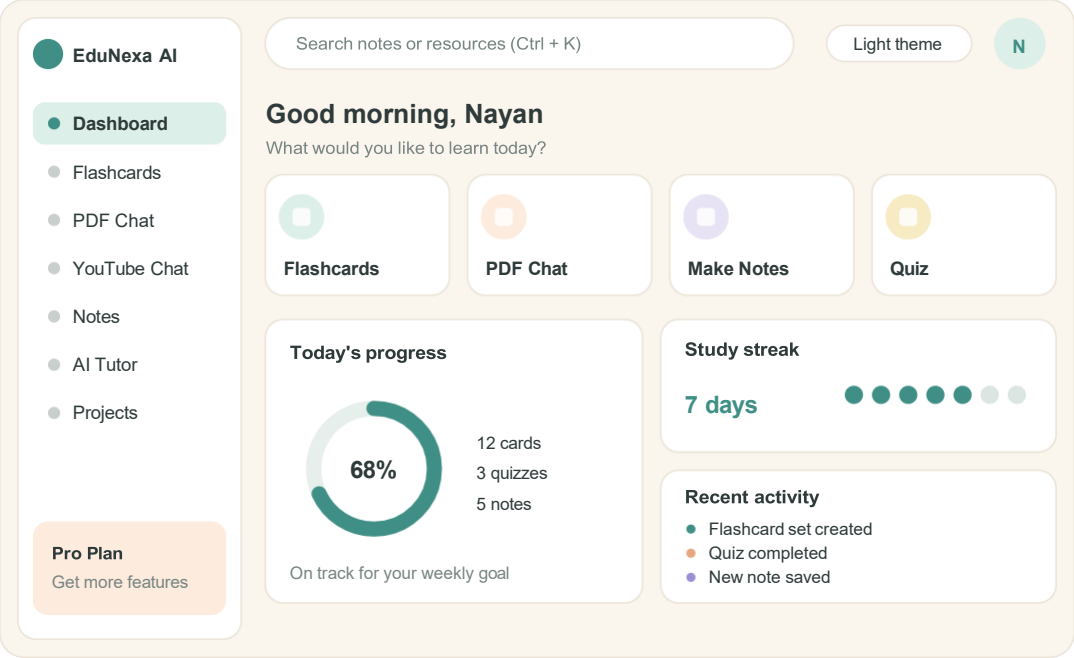


Figure 1: Proposed dashboard in the soft theme (UI mockup; numbers are sample data)

4.1 How AI is integrated

1. The uploaded PDF or video transcript is converted to text and split into small chunks.
2. Each chunk is turned into a vector (an embedding) and stored in the database with pgvector.
3. When the student asks a question, the system finds the chunks closest in meaning and sends only those to the language model (OpenAI or Gemini).
4. The model answers from those chunks, and the answer is shown together with its source text so the student can verify it.

Flashcards, quizzes, notes and project outlines use carefully designed prompts that return structured

output (JSON), which the application then displays in its own screens.

5. Technology Stack :

Frontend	Next.js (React), TypeScript, Tailwind CSS, shadcn/ui, Framer Motion, TanStack Query, Zustand, Recharts, next-themes	Pages, components, animations, data fetching, charts, light and dark theme
Backend	Node.js with Next.js API routes, Prisma ORM	APIs, business logic, rate limits, database access
Database	PostgreSQL on Supabase with the pgvector extension	Users, cards, notes, chats, projects and embeddings
Authentication and Storage	Supabase Auth (email and Google), Supabase Storage	Sign up, login, and storage of PDFs and images
AI Integration	OpenAI API (ChatGPT), Google Gemini API, LangChain, pdf-parse, youtube-transcript	Text generation, answers, RAG pipeline, PDF text and video transcripts
Payment (sandbox)	SSLCommerz or bKash sandbox	Pro plan demonstration
Deployment	Vercel, Sentry, GitHub	Hosting, error tracking and source control

6. Expected Outcome

6.1 What users will gain from the system

- One workspace for flashcards, notes, quizzes, document chat, a tutor and a project helper, instead of several separate apps.
- Answers that are based on their own course material and show their source.
- A clear view of their progress and study habit.
- A free or low-cost tool that fits a student budget.

6.2 How the system improves the existing workflow

- Making notes and revision cards, which now takes hours by hand, is reduced to a few minutes.
- Students start a course project with ideas, an outline and a report structure instead of a blank page.

- Switching between apps is removed, so more study time is spent on actual learning.

6.3 Expected benefits of AI integration

- Content-based answers (RAG) reduce wrong or invented answers and keep replies related to the syllabus.

Automatic generation of cards, questions and notes saves time and gives students practice material on demand.